

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

S V S 3 2 1

Designing your serverless API workloads for massive scale

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Principal Solutions Architect
Amazon Web Services



Agenda

What we will discuss

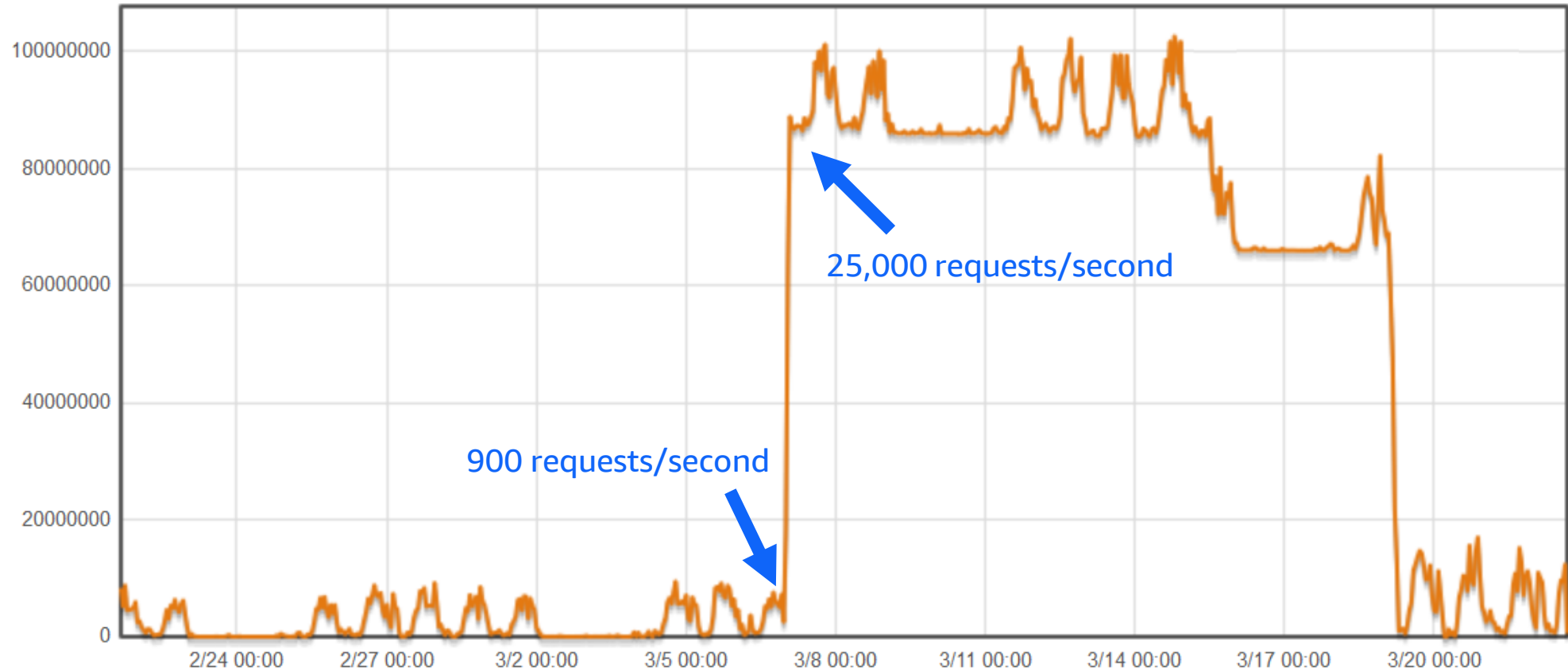
- Strategies for achieving massive scale with serverless services, including Amazon API Gateway, AWS Lambda, Amazon CloudFront, and Amazon DynamoDB
- Applied techniques with sample application
- Best practices and tools for use with your next serverless workloads

What we won't discuss

- What is serverless?
- What are Amazon API Gateway, Amazon DynamoDB, AWS Lambda?
- Other "Serverless" services (AWS Glue, OpenSearch, MSK, RedShift, etc.)

What is massive scale?

Massive scale

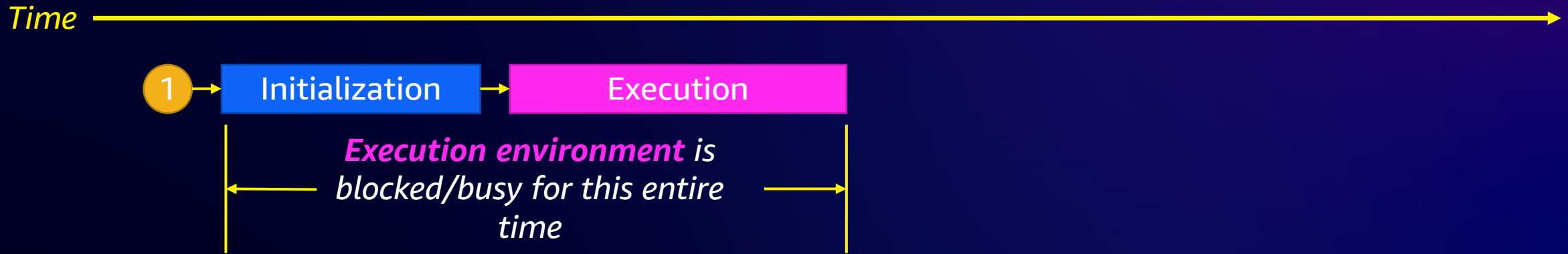


Understanding how AWS serverless API services scale

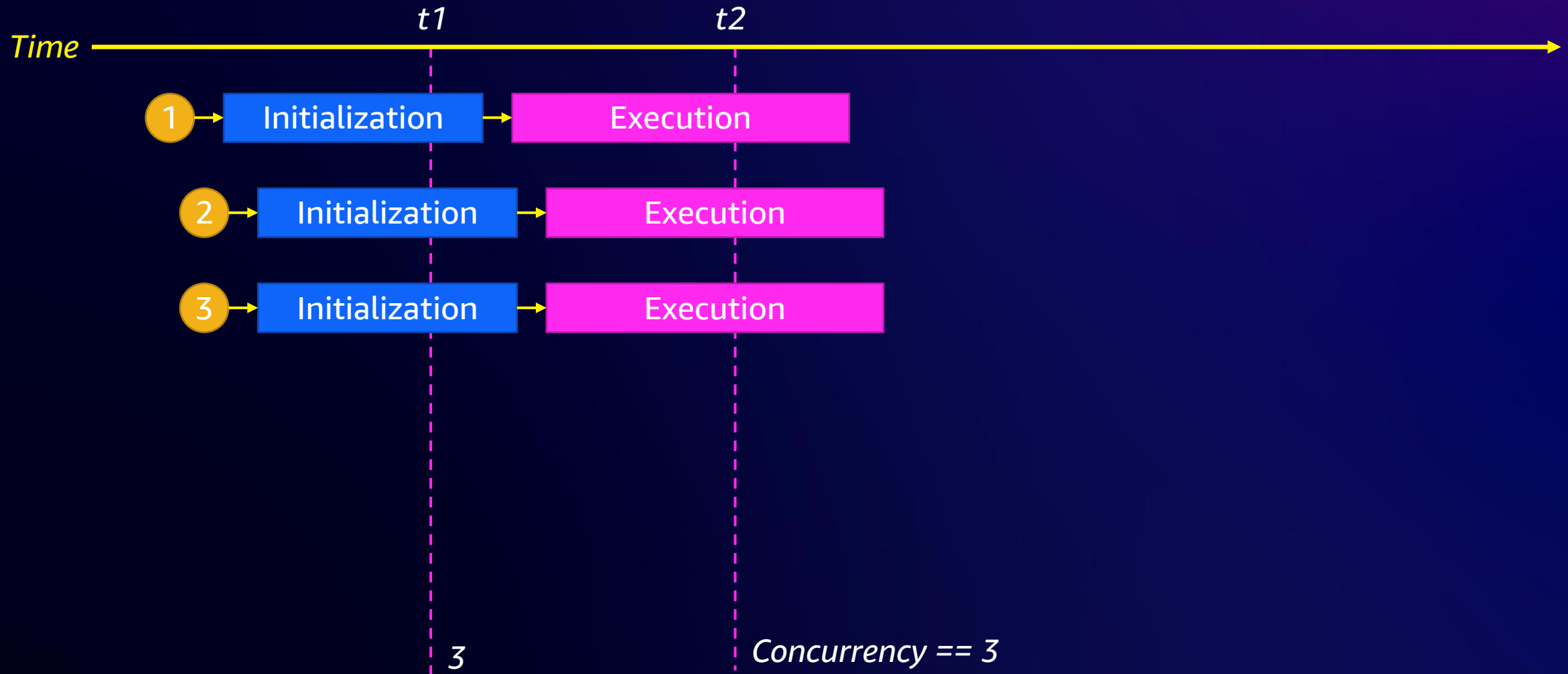
Serverless API



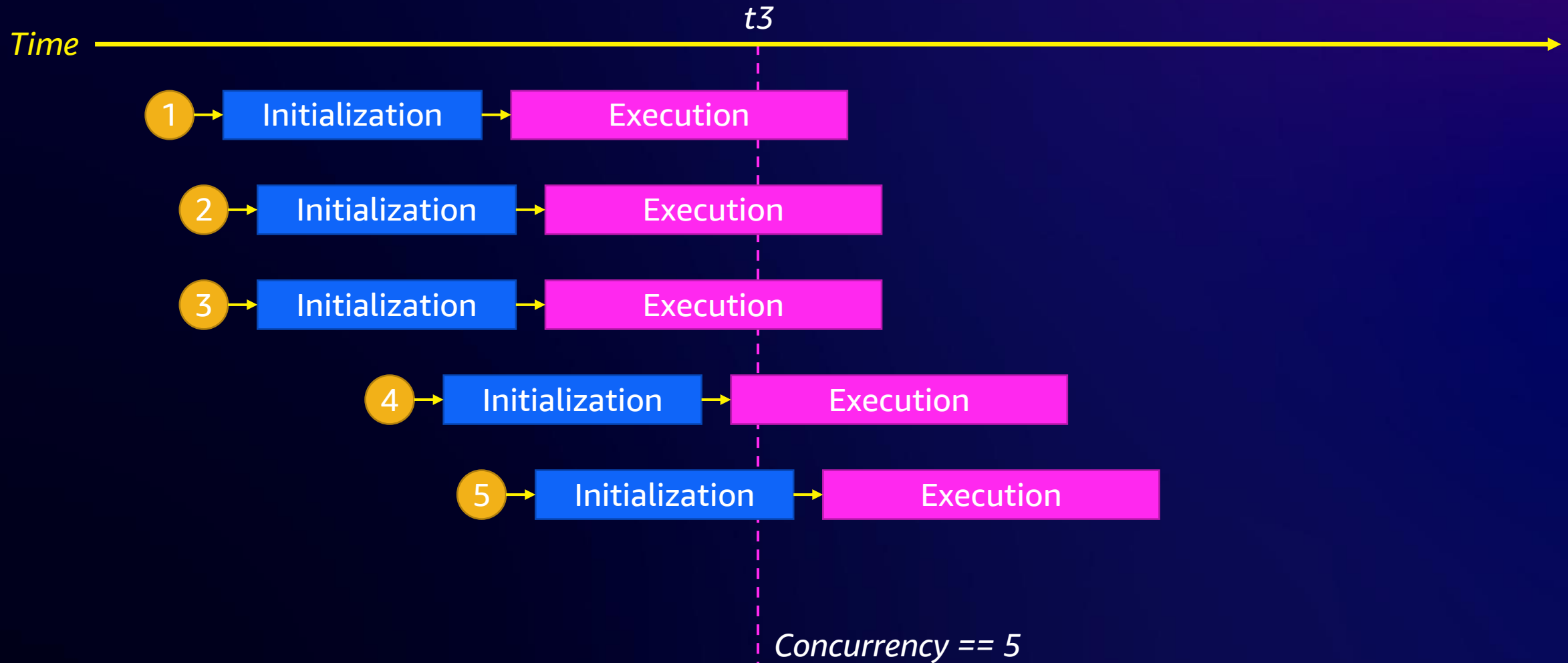
Lambda concurrency example



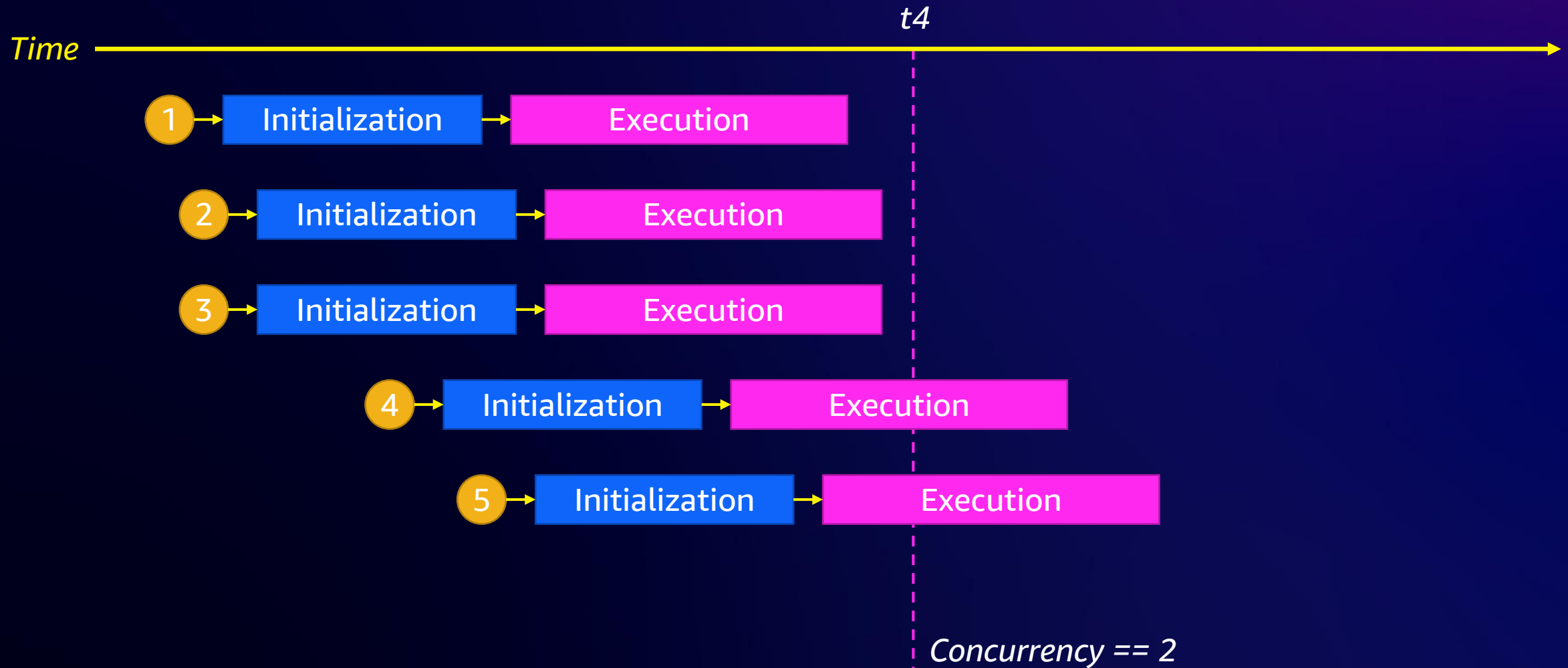
Lambda concurrency example



Lambda concurrency example



Lambda concurrency example





ServerlessVideo

A live video streaming application built with serverless technologies.



serverlessvideo

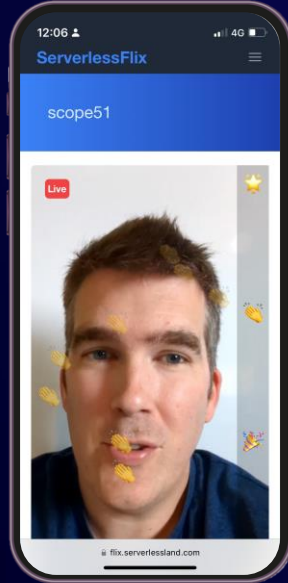
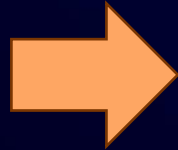
<https://s12d.com/video>

Live streaming and on-demand video

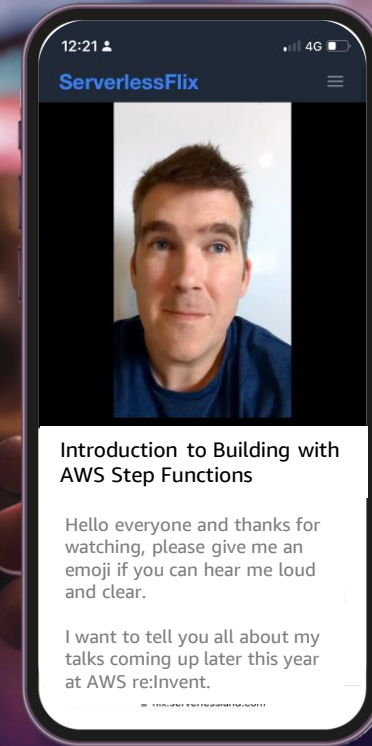
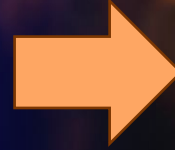
ServerlessVideo is a live video streaming application, built with serverless architecture:



Live broadcasts from experts during AWS re:Invent



Watch and interact live



Playback on demand



serverlessvideo



<https://s12d.com/video>



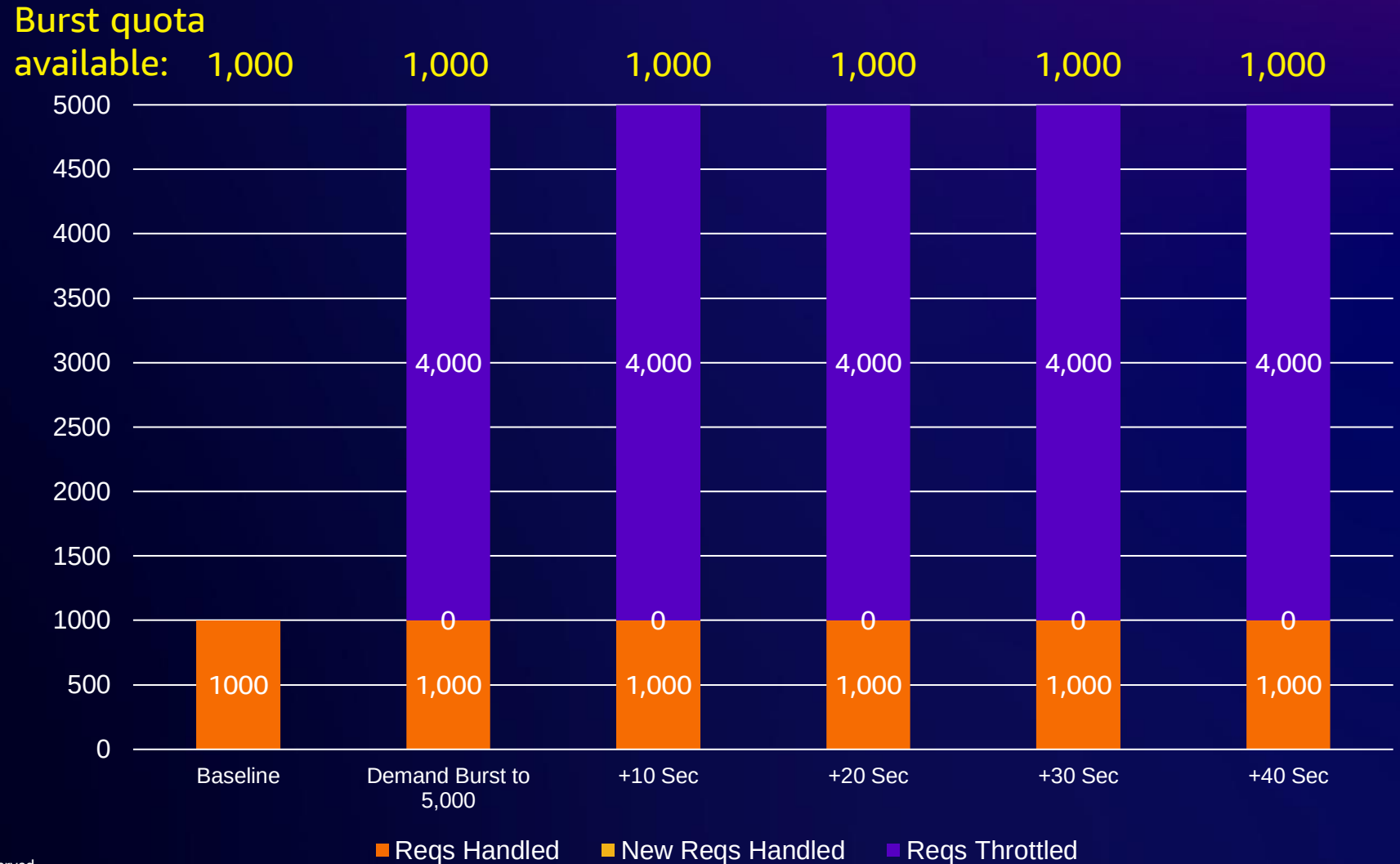
Scenario: Provisioning for peak

Lambda bursting and ramp up

Concurrency quota = 1,000 (default, us-east-1)

Concurrency quota at 1,000
default for quota

Burst to 5,000 demand
requests:
Throttled by account
concurrency quota



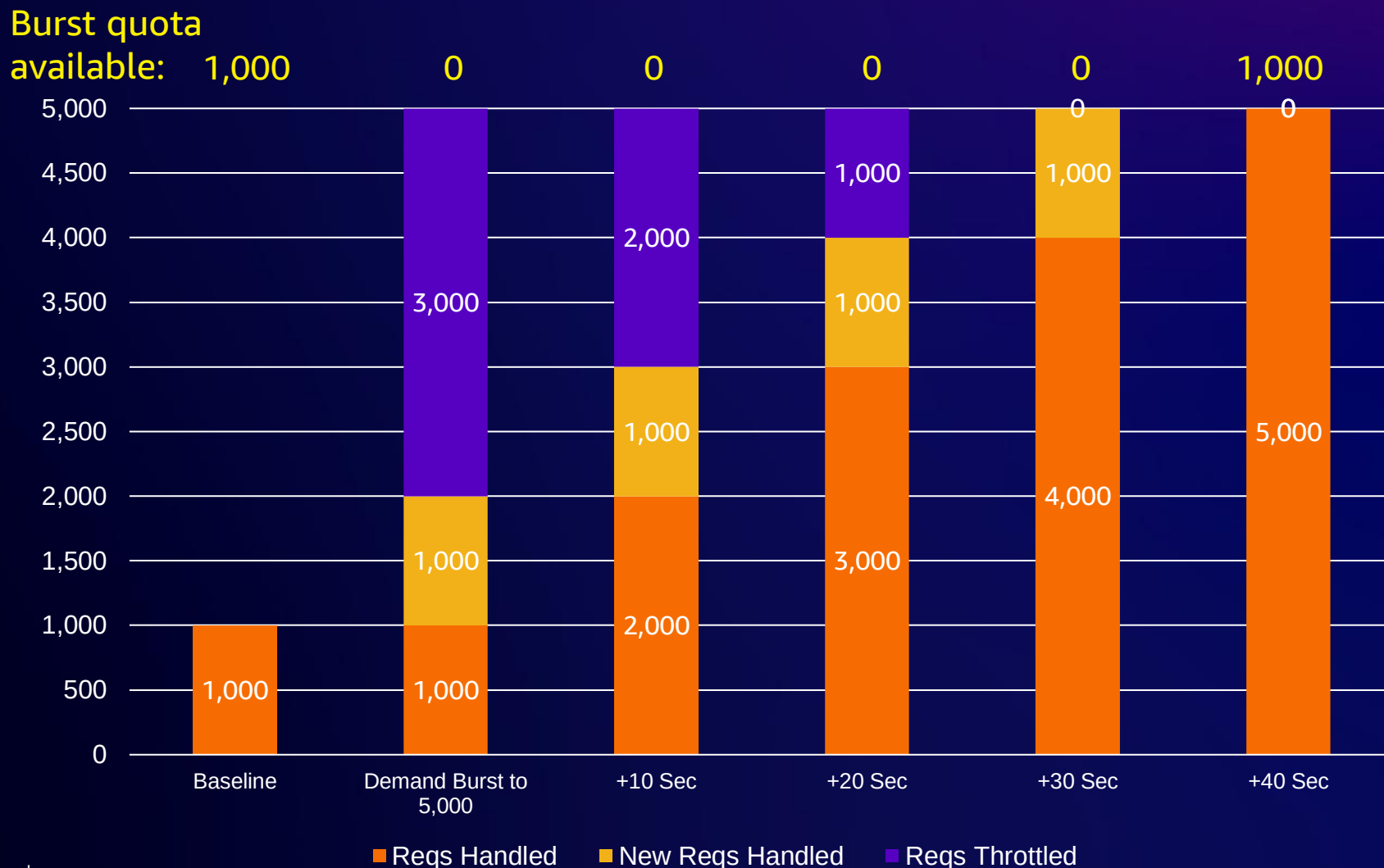
Lambda bursting and ramp up

Concurrency quota = 5,000 (increased, us-east-1)

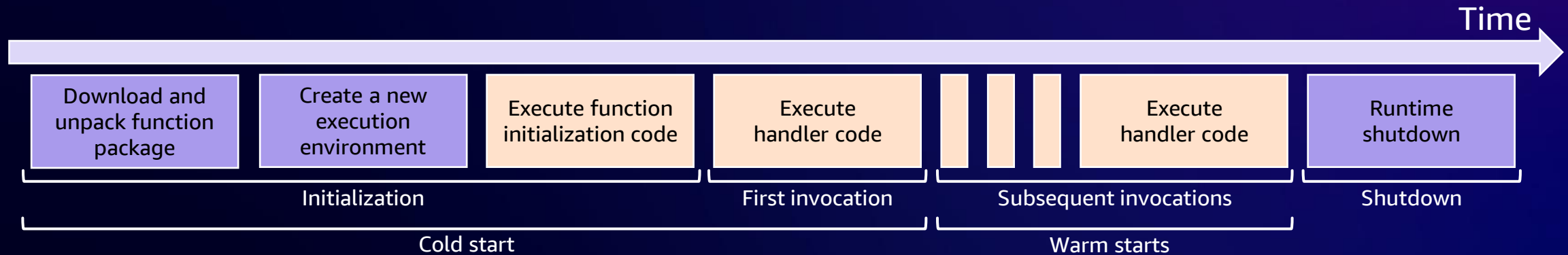
Concurrency quota increased to 5,000

Burst to 5,000 demand requests: 2,000 serviced, then reach burst quota.

After each 10 seconds, 1,000 additional concurrency until demand of 5,000 reached



Lambda execution environment lifecycle



Init: create a new or thaw existing execution environment. For new environments, run the function's initialization code.



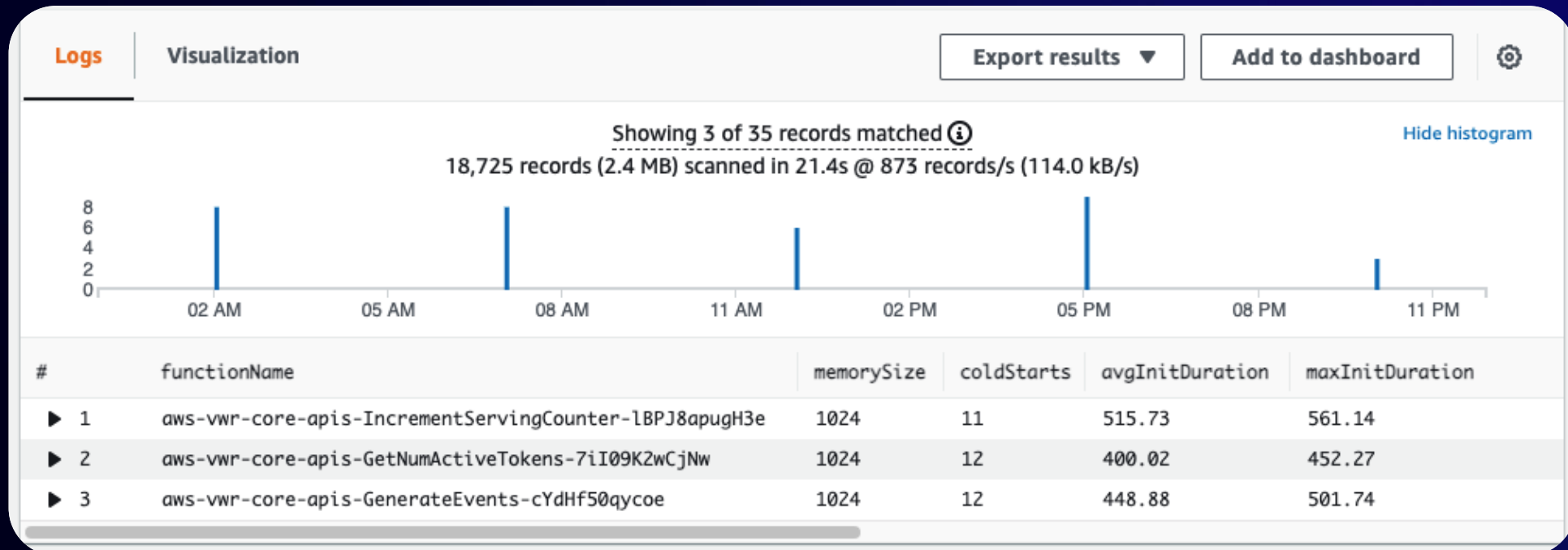
Invoke: Run the function handler code, return response. Freeze execution environment until next invocation



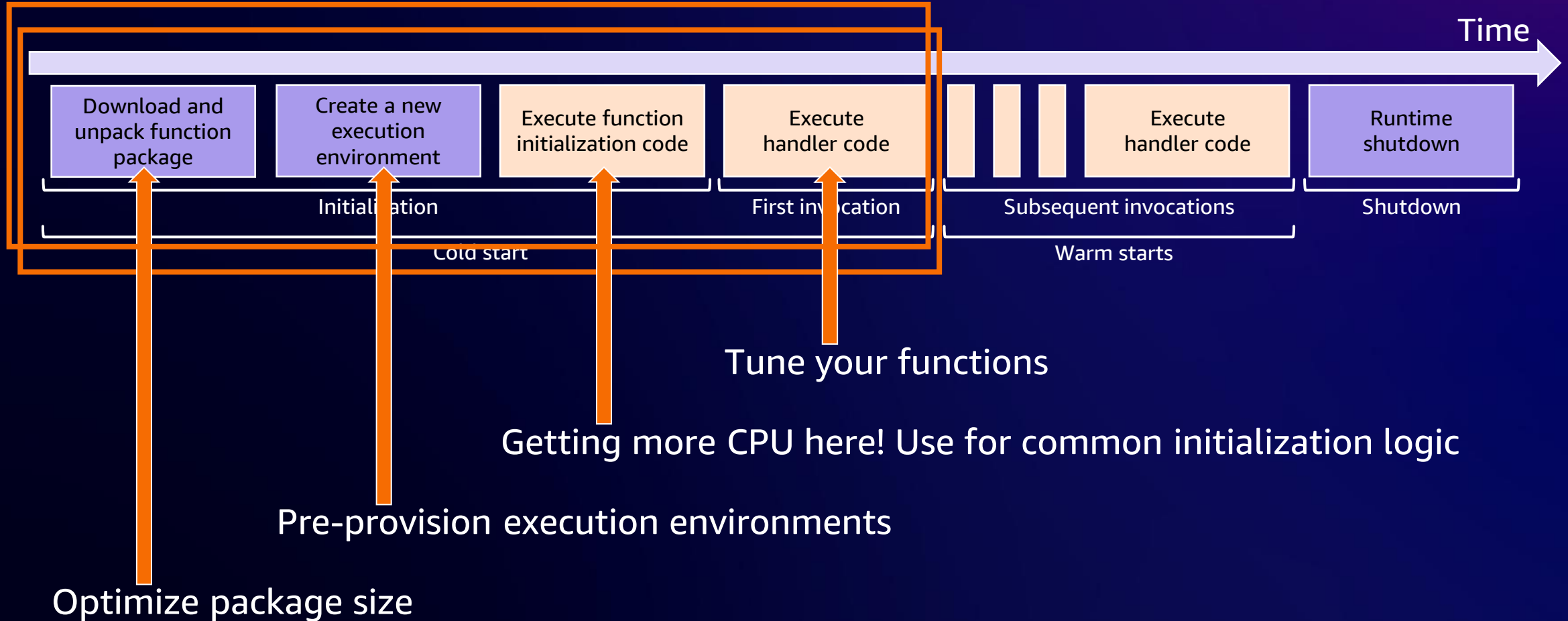
Shutdown: triggered when execution environment is destroyed, e.g. if the function does not receive invocations for a period of time

CloudWatch Logs Insights: Profiling cold starts

```
filter @type="REPORT"
| fields @memorySize / 1000000 as memorySize
| filter @message like /(?!)(Init Duration)/
| parse @message /^REPORT.*Init Duration: (?<initDuration>.*) ms.*/
| parse @log /^.*\aws\/lambda\/(?<functionName>.*)/
| stats count() as coldStarts, median(initDuration) as avgInitDuration, max(initDuration) as
maxInitDuration by functionName, memorySize
```



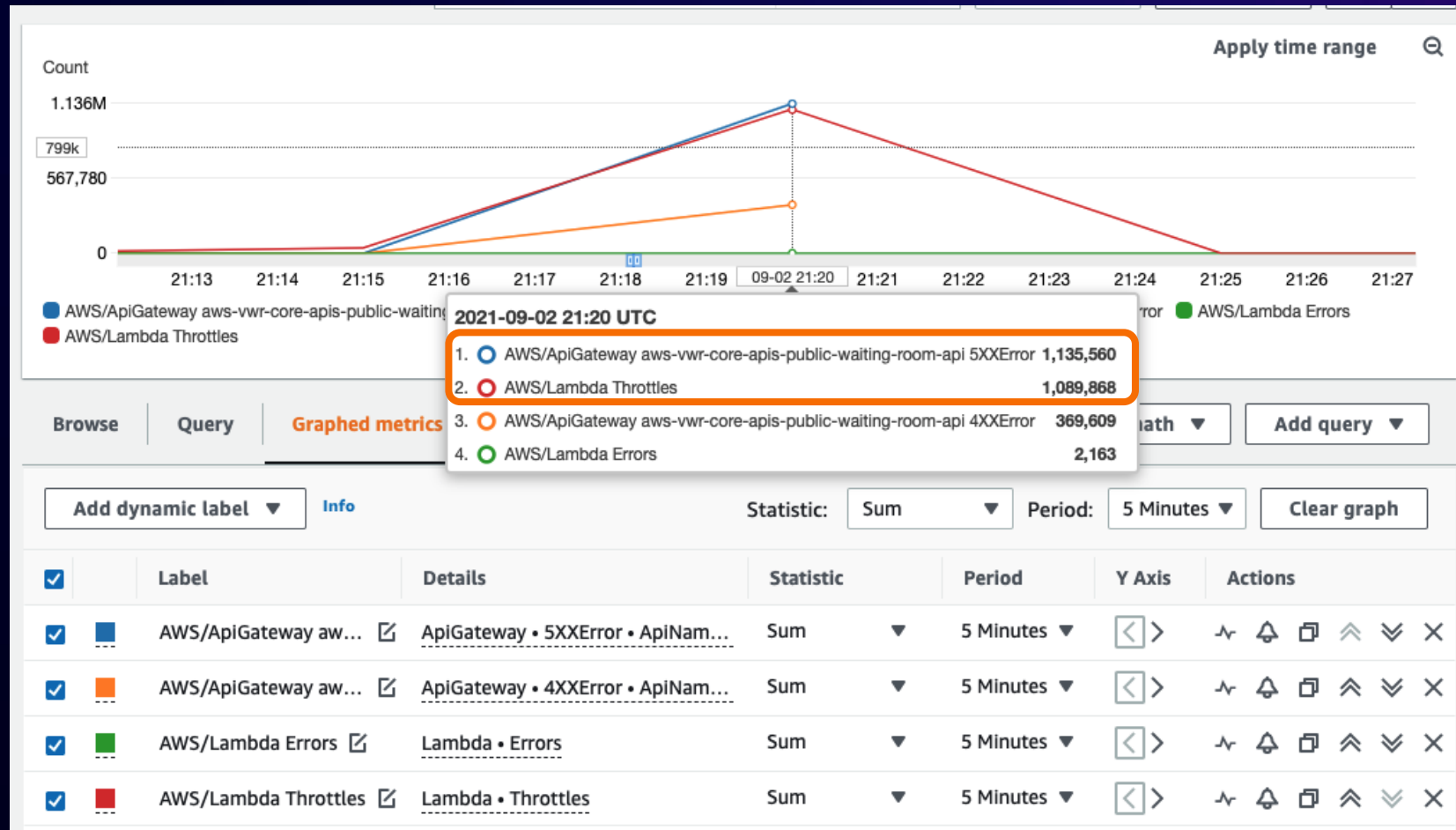
Lambda cold starts



Scenario: Scaling asynchronously

Scenario: Prepare for production

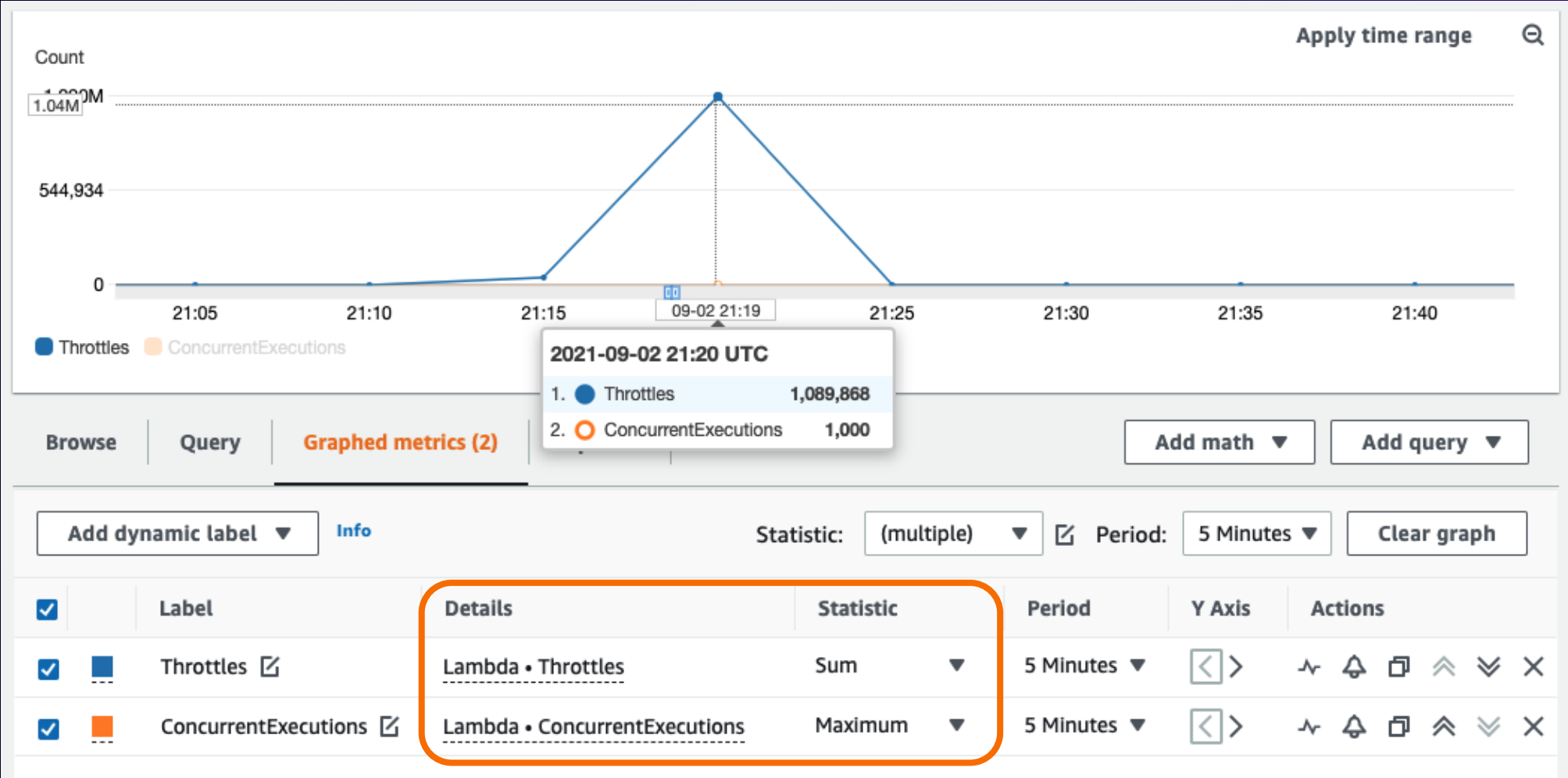
CloudWatch metrics: AWS Lambda errors/throttles



Lambda 429 TooManyRequests throttles surfaced as:

- 500 status code response from REST API Gateway deployment
- 503 status code response from HTTP API Gateway deployment

Lambda concurrency quota reached



Open discussion

Wrap up



Serverless massive scale best practices

1. Understand how serverless compute scales (service behaviors/quotas)
2. Optimize Lambda performance
3. De-couple using asynchronous processing where possible
4. Prepare your databases for massive scale
5. Leverage caching at multiple layers
6. Provision for peak when in doubt
7. Perform end-to-end load testing
8. Mitigate traffic bursts through API rate limiting

Check out these other sessions

SVS301: Building APIs: Choosing the best API solution & strategy for workloads

Tuesday, Nov. 28 at 1:00 PM – MGM Grand 115

SVS311: Optimizing cost and performance for AWS Lambda-based workloads

Wednesday, Nov. 29 at 1:00 PM – MGM 306

SVS401: Best practices for serverless developers

Thursday, Nov. 30 at 12:30 PM – Venetian Summit Showroom



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knowledge by achieving
digital badges

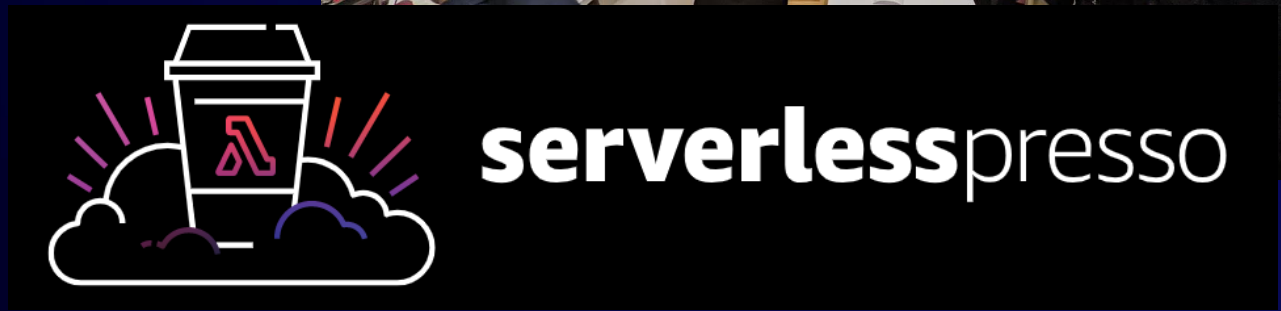
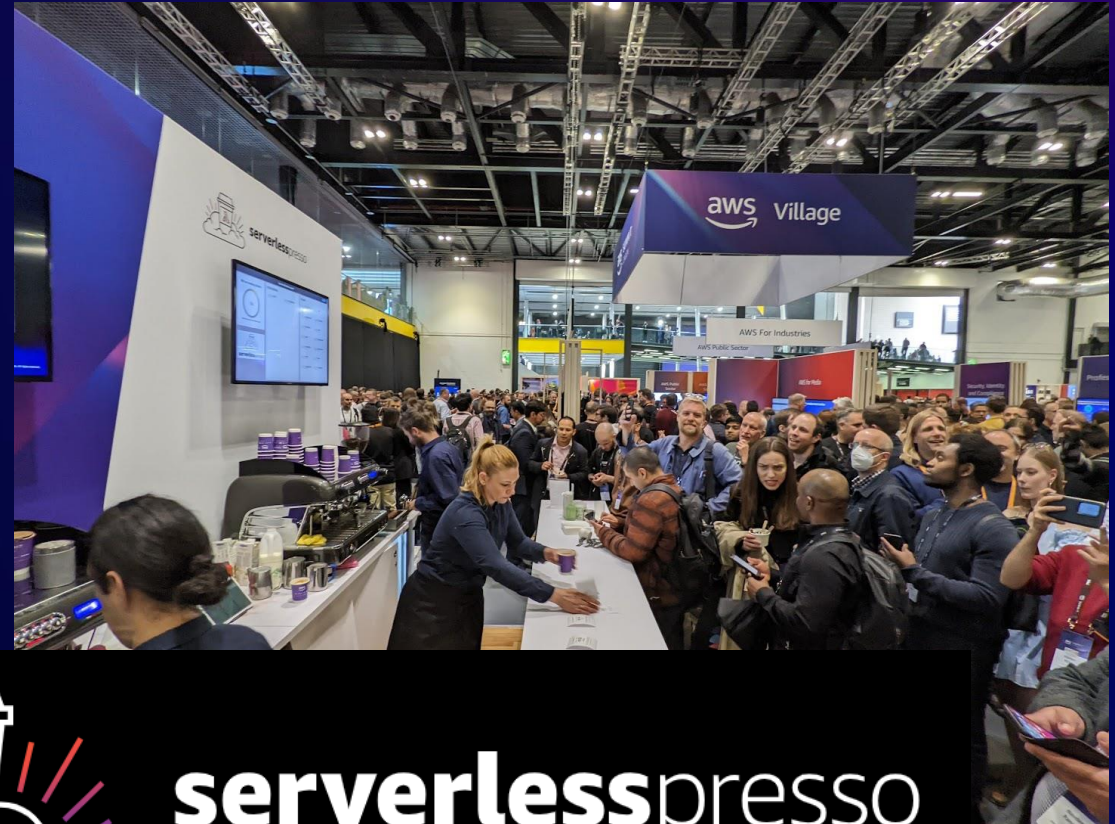


<https://s12d.com/serverless-learning>

Visit us at re:Invent Expo

Chat with experts at:

- AWS Village
- Serverlesspresso
- AWS Modern Application and Open Source Zone



Session resources



<https://s12d.com/reInvent2023-Serverless-Massive-Scale>

Thank you!



Please complete the session survey in the mobile app

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